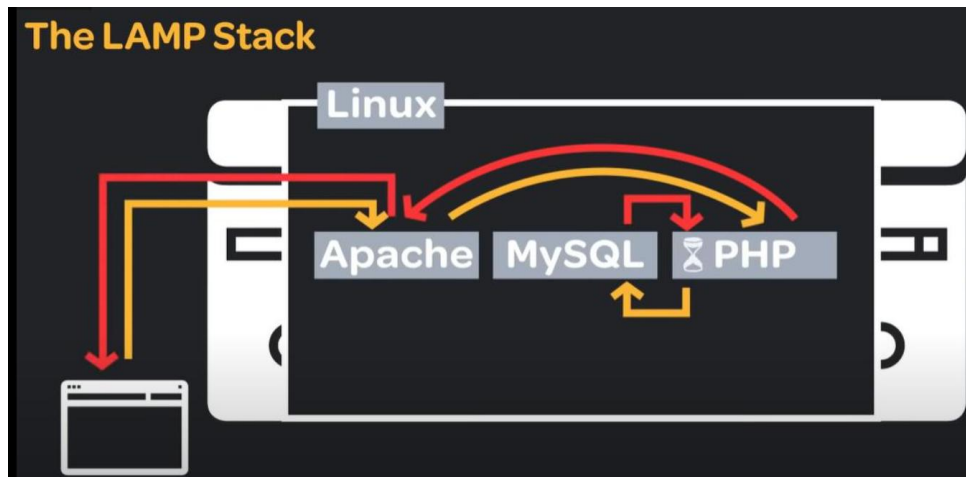


Name : Vivek Chaudhari

Project :13

Stackify : Scalable LAMP web application deployed on AWS

Architecture Diagram :



Project Objective

To deploy a real-world **LAMP (Linux, Apache, MySQL, PHP)** application on AWS using EC2 and RDS, enabling dynamic web hosting with a scalable cloud-based database.

Project Overview

This project demonstrates the deployment of a traditional LAMP stack application on AWS cloud infrastructure.

An EC2 instance is used to host the web server (Apache) and execute PHP scripts, while Amazon RDS provides a managed MySQL database service.

The application establishes a secure connection between the web server and database, allowing dynamic data interaction through PHP.

Technologies Used

- **Amazon EC2** – Web server (Linux)
- **Apache HTTP Server** – Hosting web pages
- **PHP** – Server-side scripting
- **Amazon RDS (MySQL)** – Database

- **Security Groups** – Network control
 - **SSH** – Remote access
-

Step-by-Step Implementation

1 EC2 Setup

- Launched Amazon Linux instance
- Configured security group (ports 22, 80)

2 Apache Installation

- Installed Apache using dnf
- Started and enabled service

3 PHP Configuration

- Installed PHP and PHP-FPM
- Configured Apache with PHP-FPM
- Tested using info.php

4 RDS Setup

- Created MySQL database in RDS
- Enabled public access
- Configured port 3306

5 Database Connection

- Created db.php
- Connected PHP with RDS endpoint
- Verified successful connection

lamp-db

Summary

DB Identifier

lamp-db

CPU

6.70%

Status

Available

Class

db.t4g.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

ap-south-1c

Recommend

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Tags

Reco

Connect using

Info

Code snippets

Use when connecting through SDK, API, or third-party tools including agents.

CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

Endpoints

Use when connecting through any IDE interface.

Database name

mysql

Master username

admin

Internet access gateway

Disabled

Port

3306

Endpoint type

Instance endpoint

```
Copyright (c) 2000, 2010, Oracle, MySQL Corporation AB and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE lampdb;
Query OK, 1 row affected (0.062 sec)

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| lampdb |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.012 sec)

MySQL [(none)]> exit
Bye
[ec2-user@ip-172-31-8-207 ~]$ sudo nano /var/www/html/db.php
[ec2-user@ip-172-31-8-207 ~]$
```

http://3.7.46.96/info.php

PHP Version 8.5.4	
	
System	Linux ip-172-31-8-207 ap-south-1 compute.internal 6.1.166-197.305 amzn2023.x86_64 #1 SMP PREEMPT_DYNAMIC Mon Mar 23 09:53:26 UTC 2026 x86_64
Build Date	Mar 10 2026 23:15:23
Build System	Linux
Build Provider	Amazon Linux
Compiler	gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mysqlnd.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-posix.ini, /etc/php.d/20-shmop.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sodium.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-sysmsg.ini, /etc/php.d/20-sysvsem.ini, /etc/php.d/20-sysvshm.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-mysqli.ini, /etc/php.d/30-pdo_mysqli.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmlreader.ini
PHP API	20250925
PHP Extension	20250925
Zend Extension	420250925
Zend Extension Build	API420250925.NTS
PHP Extension Build	API20250925.NTS
PHP Integer Size	64 bits
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled

http://3.7.46.96/db.php



LAMP Stack Status



Database Connected Successfully!



Server: EC2



Database: AWS RDS MySQL



Developed by Vivek Chaudhari